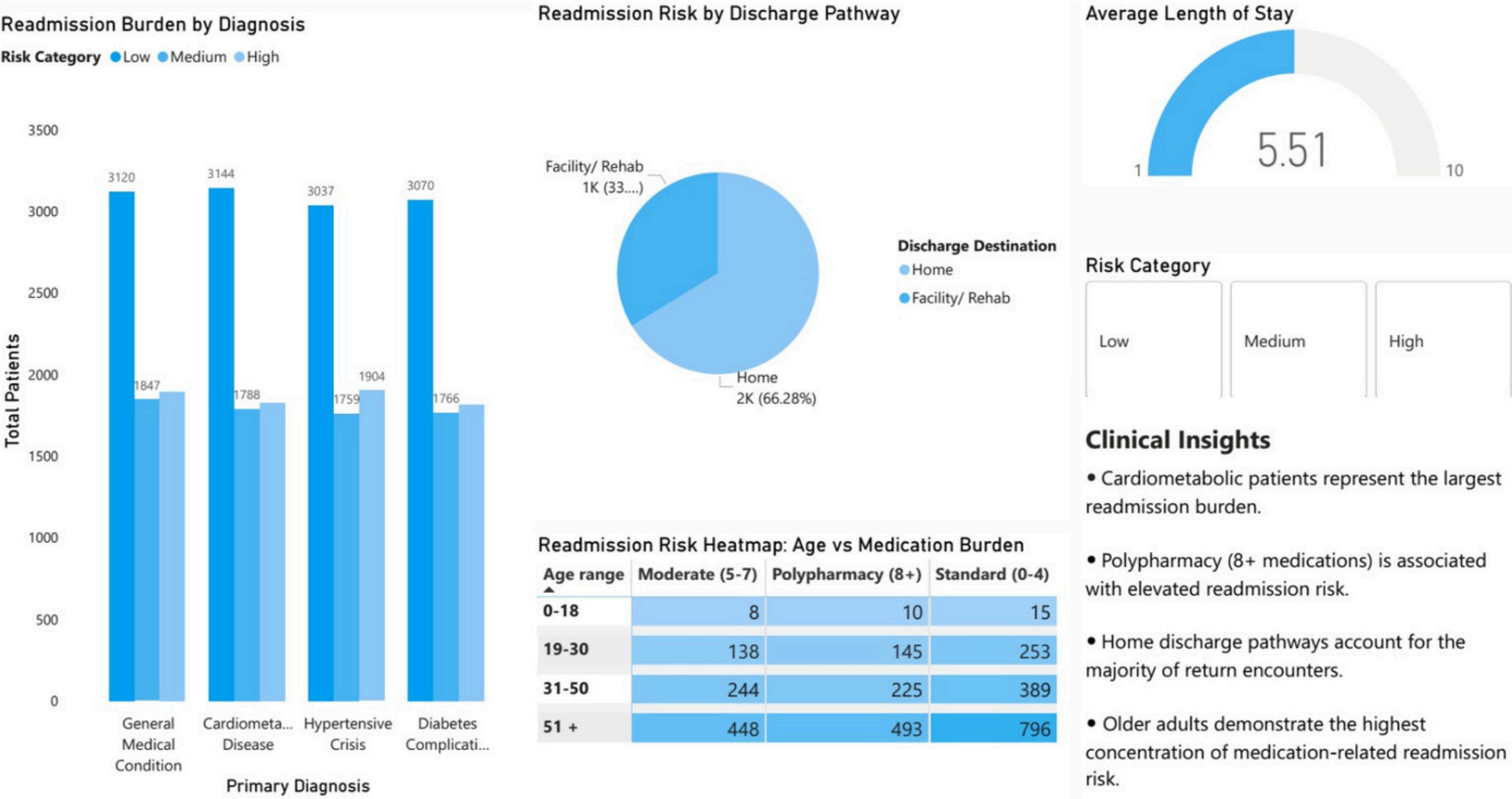


Hospital Readmission Risk Analytics Pipeline

SQL • Healthcare Analytics • Power BI

CLINICAL PERFORMANCE : 30-DAY HOSPITAL READMISSION ANALYTICS



Transforming Clinical Data into actionable Clinical & Financial Insights

SQL Development Workflow

1. Database Design

```
PROJECT: Hospital Readmission Risk Analytics
FILE: 01_database_setup.sql
OBJECTIVE:
Create foundational healthcare analytics database schema.
SKILLS:
- Database Design
- Schema Engineering
- Table Creation
*/
USE hospital_readmission_db;
CREATE TABLE hospital_encounters (
    encounter_id INT AUTO_INCREMENT PRIMARY KEY,
    patient_id INT,
    age INT,
    gender VARCHAR(20),
    blood_pressure VARCHAR(20),
    cholesterol INT,
    bmi DECIMAL(5,2),
    diabetes VARCHAR(10),
    hypertension VARCHAR(10),
    medication_count INT,
    length_of_stay INT,
    discharge_destination VARCHAR(50),
    readmitted_30_days VARCHAR(10),
    total_cost DECIMAL(12,2),
    primary_diagnosis VARCHAR(100).
```

3. Analytics Views

```
length_of_stay,
primary_diagnosis,
discharge_destination,
total_cost,
risk_category,
CASE
    WHEN readmitted_30_days = 'Yes'
    THEN 1
    ELSE 0
END AS is_readmitted
FROM hospital_encounters;
/*=====
2. FINANCIAL IMPACT VIEW
=====*/
CREATE OR REPLACE VIEW vw_financial_impact AS
SELECT
    primary_diagnosis,

    COUNT(*) AS total_cases,
    ROUND(AVG(total_cost),2) AS avg_cost_per_case,

    ROUND(SUM(total_cost),2)AS total_financial_exposure,

    ROUND(SUM(total_cost) * 0.15,2) AS
FROM hospital_encounters;
/*=====
3. CARE PATHWAYS VIEW
=====*/
CREATE OR REPLACE VIEW vw_care_pathways AS
SELECT
    discharge_destination,
    COUNT(*) AS total_patients,
    SUM(CASE
```

2. Feature Engineering

```
THEN 1500
ELSE 0
END
);

/*=====
2. PRIMARY DIAGNOSIS ENGINEERING
=====*/
UPDATE hospital_encounters
SET primary_diagnosis =
CASE
    WHEN diabetes = 'Yes'
        AND hypertension = 'Yes'
    THEN 'Cardiometabolic Disease'
    WHEN diabetes = 'Yes'
    THEN 'Diabetes Complications'
    WHEN hypertension = 'Yes'
    THEN 'Hypertensive Crisis'
    ELSE 'General Medical Condition'
END;

/*=====
3. READMISSION RISK STRATIFICATION
=====*/
USE hospital_readmission_db;
SET SQL_SAFE_UPDATES = 0;
UPDATE hospital_encounters
SET risk_category =
CASE
```

4. Quality Checks

```
Quality Assurance
- KPI Verification
*/
USE hospital_readmission_db;
/*=====
1. VERIFY RAW TABLE
=====*/
SELECT *
FROM hospital_encounters
LIMIT 10;
/*=====
2. VERIFY FINANCIAL ENGINEERING
=====*/
SELECT
    patient_id,
    total_cost,
    primary_diagnosis,
    risk_category,
    discharge_status
FROM hospital_encounters
LIMIT 20;
/*=====
3. VERIFY READMISSION VIEW
=====*/
SELECT *
FROM vw_readmission_metrics
LIMIT 10;
/*=====
4. VERIFY FINANCIAL VIEW
=====*/
SELECT *
```

Designed an end-to-end healthcare analytics database architecture using MySQL

Clinical Feature Engineering

```
USE hospital_readmission_db;
SET SQL_SAFE_UPDATES = 0;

/*=====
1. FINANCIAL COST ENGINEERING
=====*/
UPDATE hospital_encounters
SET total_cost =
(
  2500
  + (length_of_stay * 1200)
  + (medication_count * 85)
  + CASE
    WHEN bmi < 18.5
      OR bmi >= 30
    THEN 1500
    ELSE 0
  END
);

/*=====
2. PRIMARY DIAGNOSIS ENGINEERING
=====*/
UPDATE hospital_encounters
SET primary_diagnosis =
CASE
  WHEN diabetes = 'Yes'
    AND hypertension = 'Yes'
  THEN 'Cardiometabolic Disease'
  WHEN diabetes = 'Yes'
  THEN 'Diabetes Complications'
  WHEN hypertension = 'Yes'
  THEN 'Hypertensive Crisis'
  ELSE 'General Medical Condition'
END;

/*=====
3. READMISSION RISK STRATIFICATION
=====*/
USE hospital_readmission_db;
SET SQL_SAFE_UPDATES = 0;
UPDATE hospital_encounters
SET risk_category =
CASE
  WHEN medication_count >= 8 THEN 'High Risk'
  WHEN medication_count BETWEEN 5 AND 7 THEN 'Medium Risk'
  ELSE 'Low Risk'
END;

/*4. DISCHARGE OUTCOME ENGINEERING*/
UPDATE hospital_encounters
SET discharge_status =
CASE
  WHEN readmitted_30_days = 'Yes'
  THEN 'Referred Follow-up'
  ELSE 'Stable Discharge'
END;
```

- Dynamic Cost Calculation
- Diagnosis Classification
- Risk Categorization
- Clinical Feature Engineering



Transforming raw patient data into enriched analytical features

Business Intelligence Views

```
USE hospital_readmission_db;
/*=====
1. READMISSION METRICS VIEW
=====*/
CREATE OR REPLACE VIEW vw_readmission_metrics AS
SELECT
    encounter_id,
    patient_id,
    age,
    gender,
    bmi,
    diabetes,
    hypertension,
    medication_count,
    length_of_stay,
    primary_diagnosis,
    discharge_destination,
    total_cost,
    risk_category,
    CASE
        WHEN readmitted_30_days = 'Yes'
        THEN 1
        ELSE 0
    END AS is_readmitted
FROM hospital_encounters;
/*=====
2. FINANCIAL IMPACT VIEW
=====*/
CREATE OR REPLACE VIEW vw_financial_impact AS
SELECT
    primary_diagnosis,

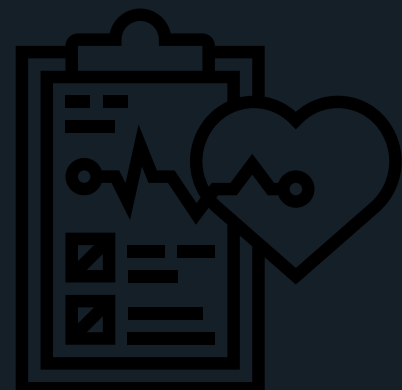
    COUNT(*) AS total_cases,
    ROUND(AVG(total_cost),2) AS avg_cost_per_case,

    ROUND(SUM(total_cost),2)AS total_financial_exposure,

    ROUND(SUM(total_cost) * 0.15,2) AS
    FROM hospital_encounters;
/*=====
3. CARE PATHWAYS VIEW
=====*/
CREATE OR REPLACE VIEW vw_care_pathways AS
SELECT
    discharge_destination,
    COUNT(*) AS total_patients,
    SUM(CASE
        WHEN readmitted_30_days = 'Yes'
        THEN 1
        ELSE 0
    END) AS total_readmissions,
    ROUND(
        100.0 *
        SUM(CASE
            WHEN readmitted_30_days = 'Yes'
            THEN 1
            ELSE 0
        END) / COUNT(*),2) AS readmission_rate_pct
FROM hospital_encounters
GROUP BY discharge_destination;
```

Skills Demonstrated

- SQL
- Healthcare Analytics
- Data Modeling
- Feature Engineering
- Data Quality Validation
- Power BI



Business Intelligence Output

CLINICAL STRATEGY & OPERATIONAL EFFICIENCY

REDUCING AVOIDABLE HOSPITAL READMISSIONS THROUGH PREDICTIVE HEALTHCARE ANALYTICS

High Risk Readmission (%)

27.59

Cost Per Readmission

\$10K

Financial Exposure

\$31.64M

Savings Opportunity

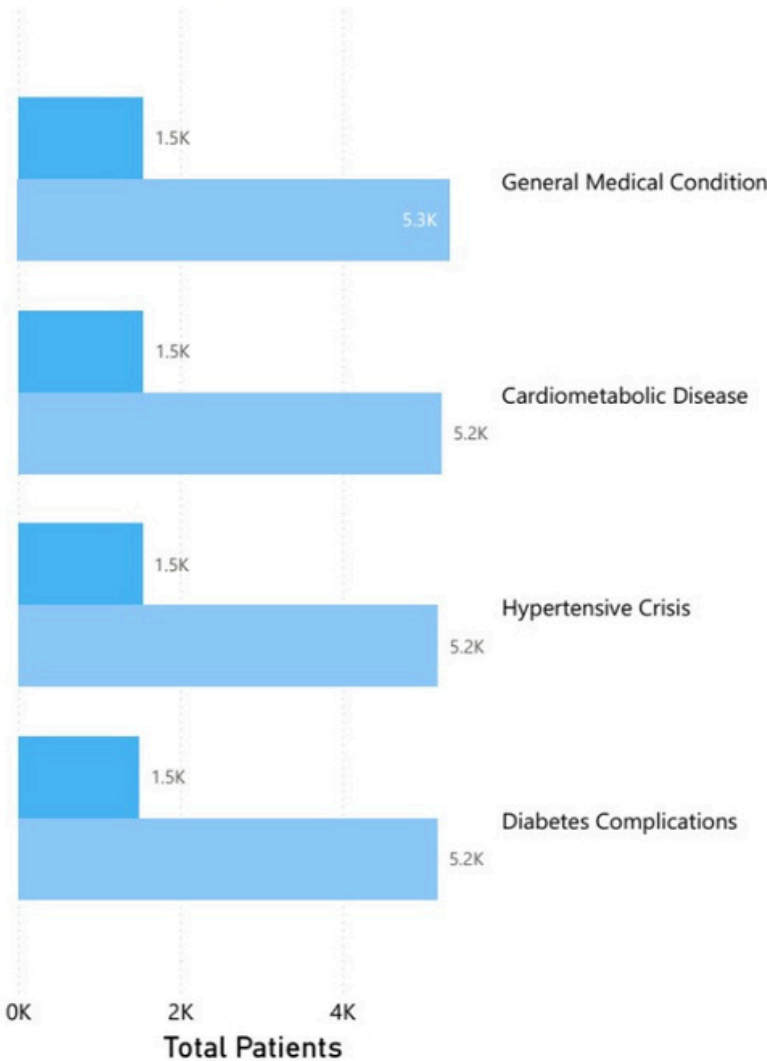
\$4.746M

Operational Burden

27.58

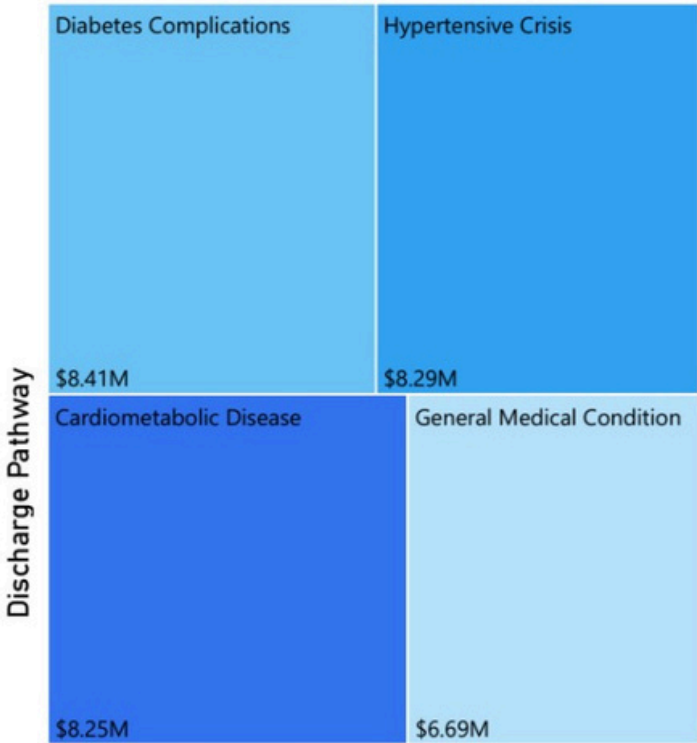
Financial Exposure by Diagnosis

Discharge Pathway ● Facility/ Rehab ● Home



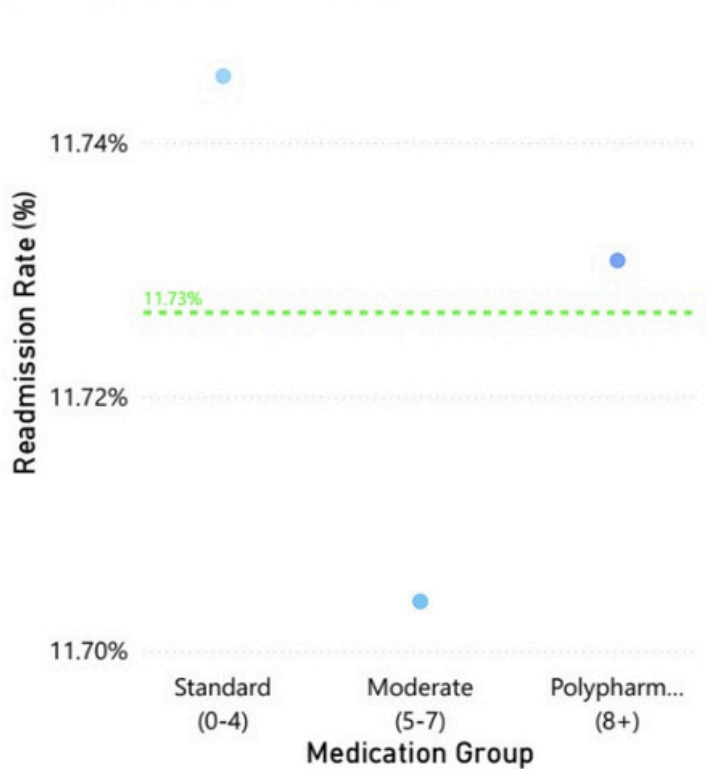
Readmission Cost Drivers

Primary Dia... ● Diabet... ● Hypert... ● Cardio... ● Gener...



Medication Burden vs Readmission Risk

Risk Category ● Low ● Medium ● High



High-Risk Intervention Priority

Risk Group	Recommended Action
High Risk	48hr post-discharge follow-up
Medium Risk	Medication reconciliation
Low Risk	Standard discharge monitoring

KEY ANALYTIC INSIGHTS

- Patients aged 51+ discharged home show elevated readmission risk.
- Polypharmacy patients demonstrate higher 30-day readmission probability.
- Early follow-up programs may reduce high-risk readmissions.



GitHub Repository Available :

<https://github.com/brianmtepe/hospital-readmission-risk-predictor>



Power BI